

Администрирование локальных сетей

Лабораторная работа №10

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Цели и задачи работы

Освоить настройку прав доступа пользователей к ресурсам сети с помощью списков управления доступом (ACL).

Выполнение лабораторной работы

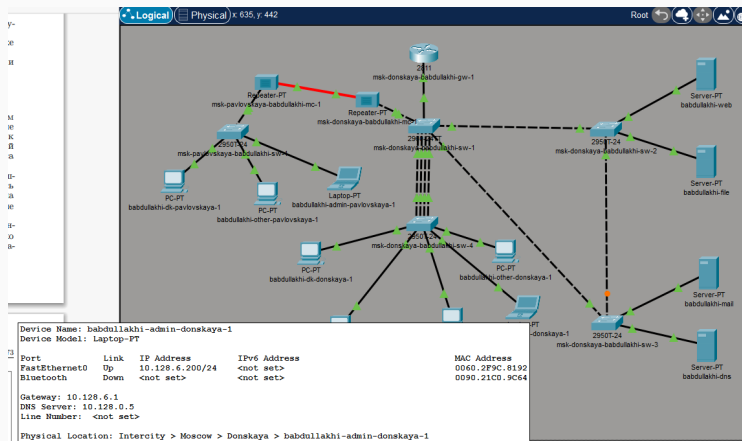


Рис. 1: Топология сети

Настройка IP-адреса администратора

ть действия установленных правил доступа
самостоятельной работы по настройке
на Пасковской.

аполнения работы

хотите ноутбук администратора с именем
тем, чтобы разрешить ему потом любые
сетью. Для этого подсоедиайте ноутбук
кава-4 и присвойте ему статический
the gateway-адреса 10.128.6.1 и адреса

(см. рис. 9.2) будем настраивать на маршрутизаторе

Device Name: babdullakhi-admin-pavlovskaya-1

Device Model: Laptop-PT

Port	Link	IP Address	IPv6 Address	MAC Address
FastEthernet0	Up	10.128.6.222/24	<not set>	00E0.F996.E42A
Bluetooth	Down	<not set>	<not set>	00E0.5000.4B55

Gateway: 10.128.6.1

DNS Server: 10.128.0.5

Line Number: <not set>

Physical Location: InterCity > Moscow > Pavlovskaya > babdullakhi-admin-pavlovskaya-1

ОБЗОР ЛОКАЛЬНЫХ СИСТЕМ

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Москва, Россия

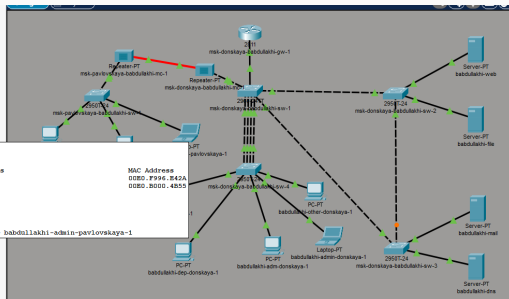


Рис. 2: Настройка IP-адреса администратора

```
msk-donskaya-babdullakhi-gw-1>en
Password:
msk-donskaya-babdullakhi-gw-1#conf t
Enter configuration commands, one per line. End with CNTL/Z.
msk-donskaya-babdullakhi-gw-1(config)#ip access-list extended servers-out
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark web
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.2 eq 80
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#exit
msk-donskaya-babdullakhi-gw-1(config)#int f0/0.3
msk-donskaya-babdullakhi-gw-1(config-subif)#ip access-group servers-out out
msk-donskaya-babdullakhi-gw-1(config-subif)#exit
msk-donskaya-babdullakhi-gw-1(config)#
```

Рис. 3: Создание ACL для web-доступа

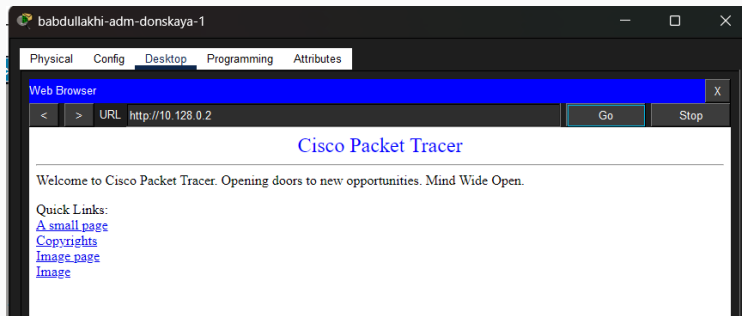
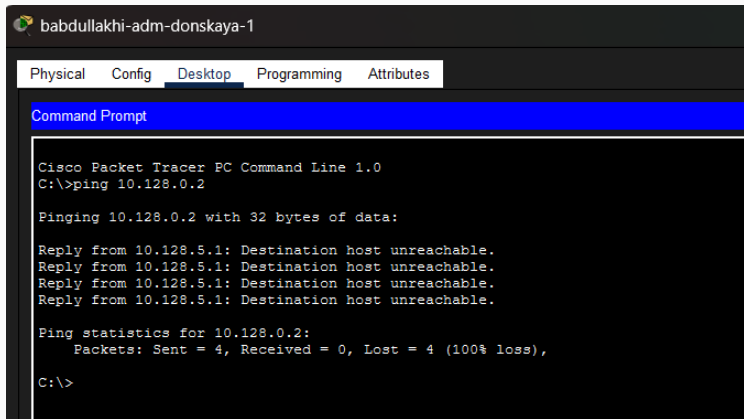


Рис. 4: Применение ACL к интерфейсу

Проверка HTTP-доступа



The screenshot shows the Cisco Packet Tracer interface with the 'Desktop' tab selected. A 'Command Prompt' window is open, displaying the output of a ping command. The window title is 'babdullakhi-adm-donskaya-1'. The command prompt shows the following text:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.128.0.2

Pinging 10.128.0.2 with 32 bytes of data:

Reply from 10.128.5.1: Destination host unreachable.
Reply from 10.128.5.1: Destination host unreachable.
Reply from 10.128.5.1: Destination host unreachable.
Reply from 10.128.5.1: Destination host unreachable.

Ping statistics for 10.128.0.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

Рис. 5: Проверка HTTP-доступа

```
msk-donskaya-babdullakhi-gw-1(config)#ip access-list extended servers-out
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp host 10.128.6.200 host 10.128.0.2 range 20 ftp
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp host 10.128.6.200 host 10.128.0.2 eq telnet
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp host 10.128.6.222 host 10.128.0.2 range 20 ftp
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp host 10.128.6.222 host 10.128.0.2 eq telnet
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#
```

Рис. 6: Проверка ping до сервера

Добавление правил для администратора

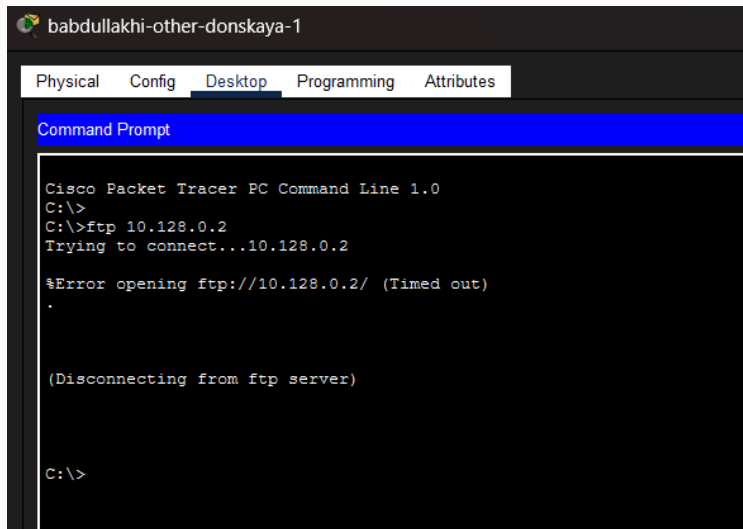
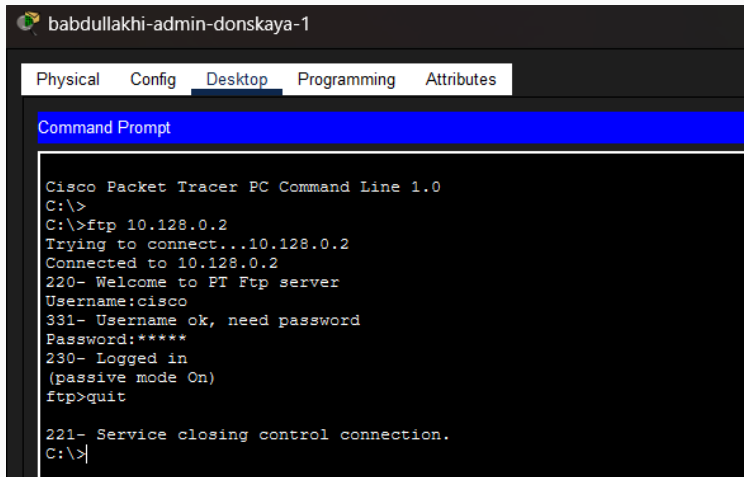


Рис. 7: Добавление правил для администратора

Проверка FTP-доступа администратора



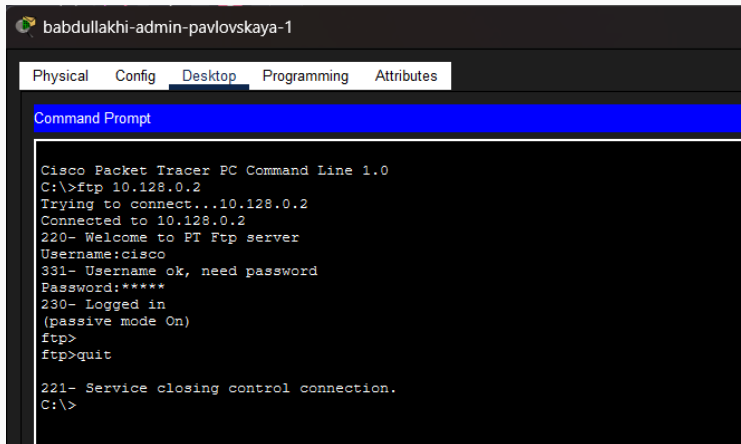
The screenshot shows a Cisco Packet Tracer interface for a device named 'babdullakhi-admin-donskaya-1'. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The command prompt shows the execution of the 'ftp' command to connect to 10.128.0.2. The connection is successful, and the user 'cisco' is logged in. The prompt then shows 'ftp>quit', and the service closing message '221- Service closing control connection.' is displayed, followed by the return to the 'C:\>' prompt.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>
C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>quit

221- Service closing control connection.
C:\>|
```

Рис. 8: Успешное FTP-подключение администратора

Ограничение FTP для других пользователей



The screenshot shows a Cisco Packet Tracer PC Command Line window for a device named 'babdullakhi-admin-pavlovskaya-1'. The 'Desktop' tab is selected. The command prompt shows a successful FTP session initiated from the local host (C:\) to 10.128.0.2 using the 'cisco' username and password. The session ends with the 'quit' command, and the control connection is closed.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ftp 10.128.0.2
Trying to connect...10.128.0.2
Connected to 10.128.0.2
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>
ftp>quit

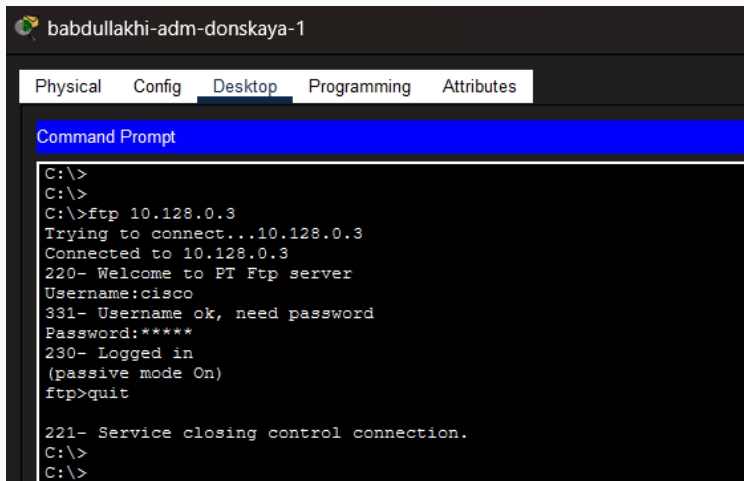
221- Service closing control connection.
C:\>
```

Рис. 9: Запрет FTP-доступа для других пользователей

```
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#  
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark file  
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445  
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.3 range 20 ftp  
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#
```

Рис. 10: Настройка ACL для файлового сервера

Проверка FTP-доступа к файловому серверу



The screenshot shows a Windows desktop environment. At the top, the taskbar displays the username 'babdullakhi-adm-donskaya-1'. Below the taskbar, there are five tabs: 'Physical', 'Config', 'Desktop' (which is selected), 'Programming', and 'Attributes'. The main area of the desktop is occupied by a 'Command Prompt' window. The window has a blue title bar and a black background with white text. The text in the Command Prompt shows the execution of an FTP command from the C:\ directory to connect to 10.128.0.3. The connection is successful, and the user 'cisco' is logged in. The user then enters 'quit' to end the session, and the service closes the control connection.

```
C:\>
C:\>
C:\>ftp 10.128.0.3
Trying to connect...10.128.0.3
Connected to 10.128.0.3
220- Welcome to PT Ftp server
Username:cisco
331- Username ok, need password
Password:*****
230- Logged in
(passive mode On)
ftp>quit

221- Service closing control connection.
C:\>
C:\>
```

Рис. 11: Проверка FTP-доступа к файловому серверу

```
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.5 range 25 105
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark mail
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 eq smtp
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit tcp any host 10.128.0.4 eq pop3
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#remark dns
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#permit udp 10.128.0.0 0.0.255.255 host 10.128.0.5 eq 53
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#1 permit icmp any any
msk-donskaya-babdullakhi-gw-1(config-ext-nacl)#exit
msk-donskaya-babdullakhi-gw-1(config)#
msk-donskaya-babdullakhi-gw-1(config)#
```

Рис. 12: Настройка ACL для mail, DNS и ICMP

Проверка доступа к web-серверу по имени

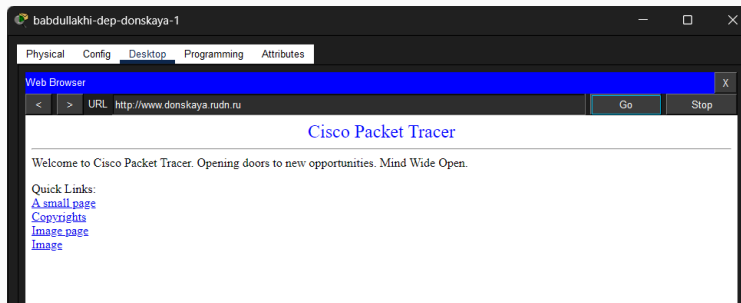


Рис. 13: Проверка доступа к web-серверу по имени

Проверка работы почтового сервера

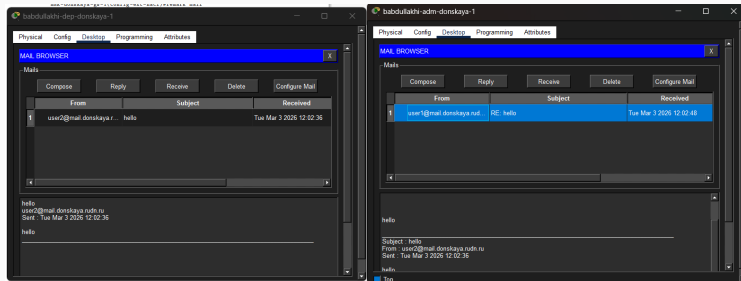
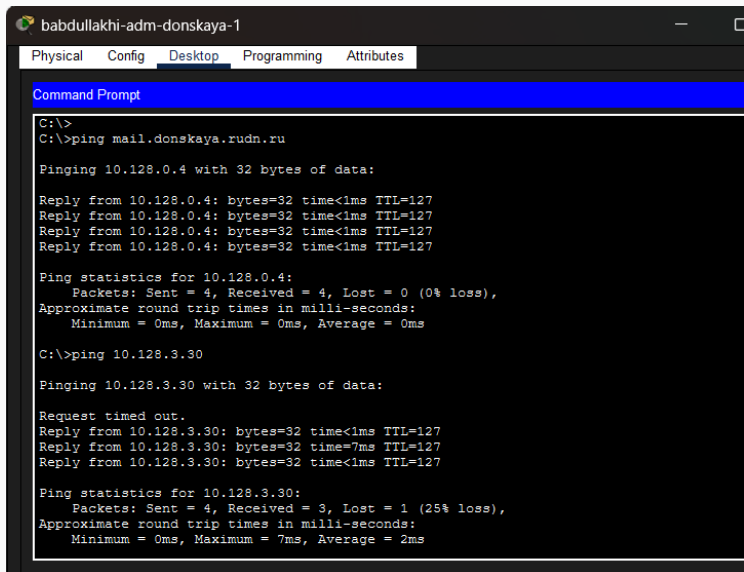


Рис. 14: Проверка работы почтового сервера

Проверка ICMP после изменения ACL



The screenshot shows a Windows Command Prompt window with the title bar 'babdullakhi-adm-donskaya-1'. The window has tabs for 'Physical', 'Config', 'Desktop', 'Programming', and 'Attributes', with 'Desktop' currently selected. The Command Prompt displays the following text:

```
C:\>
C:\>ping mail.donskaya.rudn.ru

Pinging 10.128.0.4 with 32 bytes of data:

Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127
Reply from 10.128.0.4: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

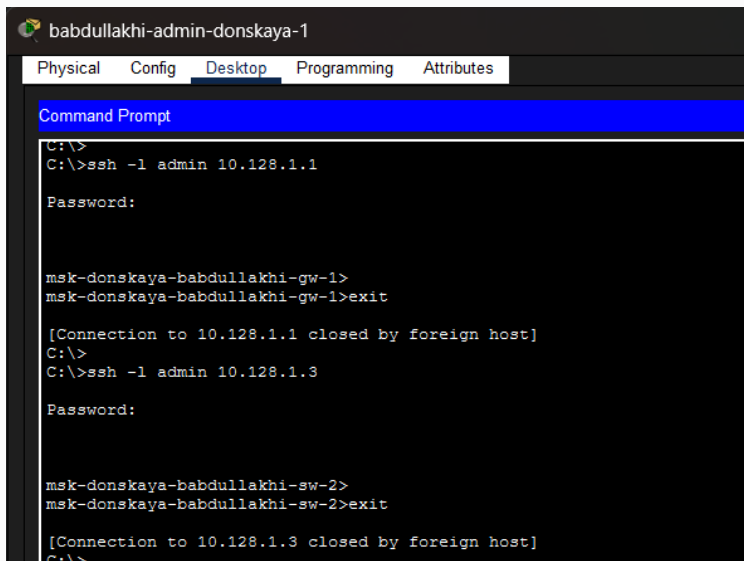
C:\>ping 10.128.3.30

Pinging 10.128.3.30 with 32 bytes of data:

Request timed out.
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127
Reply from 10.128.3.30: bytes=32 time=7ms TTL=127
Reply from 10.128.3.30: bytes=32 time<1ms TTL=127

Ping statistics for 10.128.3.30:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 7ms, Average = 2ms
```

Проверка SSH-доступа к оборудованию



```
babdullakhi-admin-donskaya-1
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>ssh -l admin 10.128.1.1

Password:

msk-donskaya-babdullakhi-gw-1>
msk-donskaya-babdullakhi-gw-1>exit

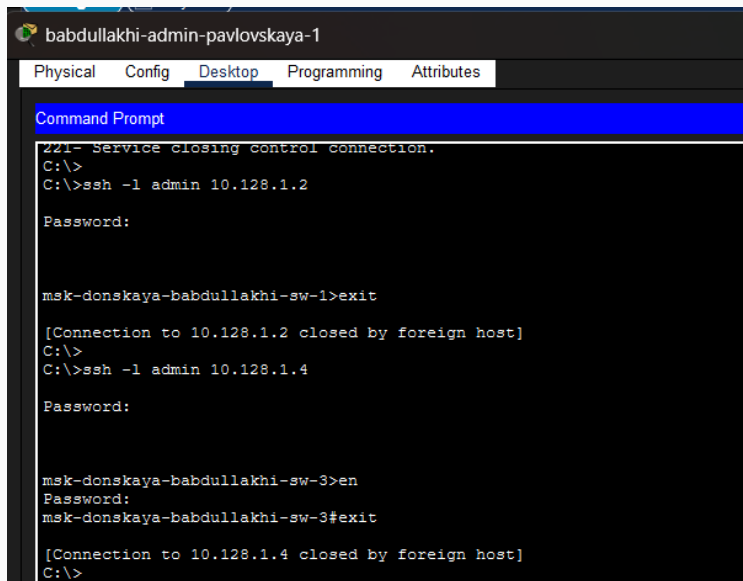
[Connection to 10.128.1.1 closed by foreign host]
C:\>
C:\>ssh -l admin 10.128.1.3

Password:

msk-donskaya-babdullakhi-sw-2>
msk-donskaya-babdullakhi-sw-2>exit

[Connection to 10.128.1.3 closed by foreign host]
C:\>
```

Проверка SSH-доступа к другим коммутаторам



The screenshot shows a network management application window titled "babdullakhi-admin-pavlovskaya-1". It has a tabbed interface with "Physical", "Config", "Desktop", "Programming", and "Attributes" tabs. The "Desktop" tab is active, displaying a "Command Prompt" window. The Command Prompt shows the following sequence of commands and outputs:

```
221- Service closing control connection.  
C:\>  
C:\>ssh -l admin 10.128.1.2  
  
Password:  
  
msk-donskaya-babdullakhi-sw-1>exit  
  
[Connection to 10.128.1.2 closed by foreign host]  
C:\>  
C:\>ssh -l admin 10.128.1.4  
  
Password:  
  
msk-donskaya-babdullakhi-sw-3>en  
Password:  
msk-donskaya-babdullakhi-sw-3#exit  
  
[Connection to 10.128.1.4 closed by foreign host]  
C:\>
```

```
msk-donskaya-babdullakhi-gw-1#sh access-lists
Extended IP access list servers-out
 1 permit icmp any any (4 match(es))
10 permit tcp any host 10.128.0.2 eq www (11 match(es))
20 permit tcp host 10.128.6.200 host 10.128.0.2 range 20 ftp (10 match(es))
30 permit tcp host 10.128.6.200 host 10.128.0.2 eq telnet
40 permit tcp host 10.128.6.222 host 10.128.0.2 range 20 ftp (10 match(es))
50 permit tcp host 10.128.6.222 host 10.128.0.2 eq telnet
60 permit tcp 10.128.0.0 0.0.255.255 host 10.128.0.3 eq 445
70 permit tcp any host 10.128.0.3 range 20 ftp (11 match(es))
80 permit tcp any host 10.128.0.4 eq smtp (11 match(es))
90 permit tcp any host 10.128.0.4 eq pop3 (10 match(es))
100 permit udp 10.128.0.0 0.0.255.255 host 10.128.0.5 eq domain (7 match(es))
Extended IP access list other-in
10 permit ip host 10.128.6.200 any (62 match(es))
20 permit ip host 10.128.6.222 any (75 match(es))
Extended IP access list management-out
10 permit ip host 10.128.6.200 10.128.1.0 0.0.0.255 (31 match(es))
20 permit ip host 10.128.6.222 10.128.1.0 0.0.0.255 (75 match(es))

msk-donskaya-babdullakhi-gw-1#
```

Рис. 18: Просмотр итоговых списков доступа

Выводы по проделанной работе

В ходе лабораторной работы была выполнена настройка списков контроля доступа **ACL** на маршрутизаторе в среде **Cisco Packet Tracer**. Были настроены правила фильтрации для web-, файлового, почтового и DNS-сервисов, а также разграничен административный доступ к сетевым ресурсам и оборудованию.

Проверка показала, что доступ разрешается или запрещается в соответствии с заданной политикой безопасности, а значит настройка ACL выполнена корректно.